

Forecasting Solar Energetic Particles (SEP) Using HOPEs and Multimodal Neural Networks

Background

Solar Energetic Particle (SEP) events happen when high-energy particles are released from the Sun into space during powerful solar activity, such as flares or coronal mass ejections. SEPs have the potential to (Hosseinzadeh et al., 2024):

- damage **satellites**
- affect **communications**
- increase **radiation** risk for **astronauts**

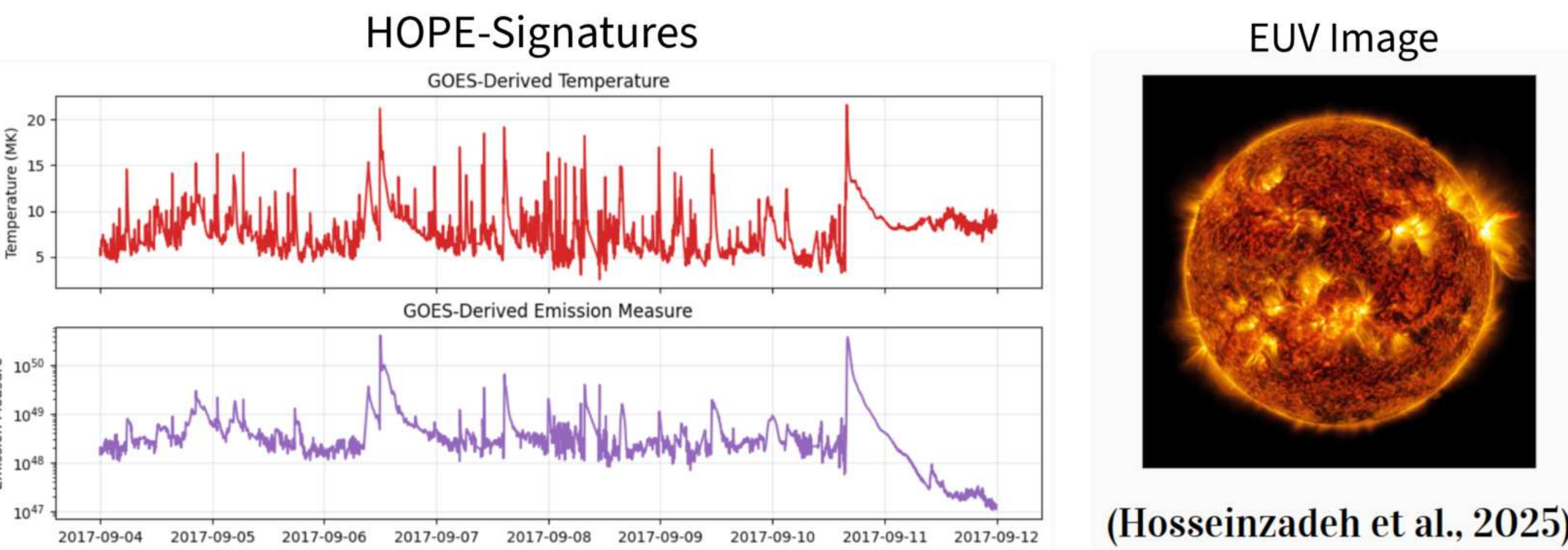
Forecasting SEPs is difficult because warning signs are subtle and SEP events are less common than non-SEP periods. Because SEP events can happen quickly, scientists need ways to ways to detect **early warning signs** (Palmerio et al., 2024). Scientists use GOES satellites to collect solar X-ray data to help scientists monitor solar activity. X-ray data can be used to estimate plasma temperature and emission measure. It can be used to estimate:

- Plasma Temperature
- Emission Measure
- Heating Rates
- Changes before SEP Events

Heat Onset Precursor Events (HOPEs) are early-phase solar flare signatures that happen 5-15 minutes before the main SEP peak.

EUV Images show the visual changes in the Earth's atmosphere.

Purpose



This project tests whether a two-branch machine learning model can improve SEP prediction by combining:

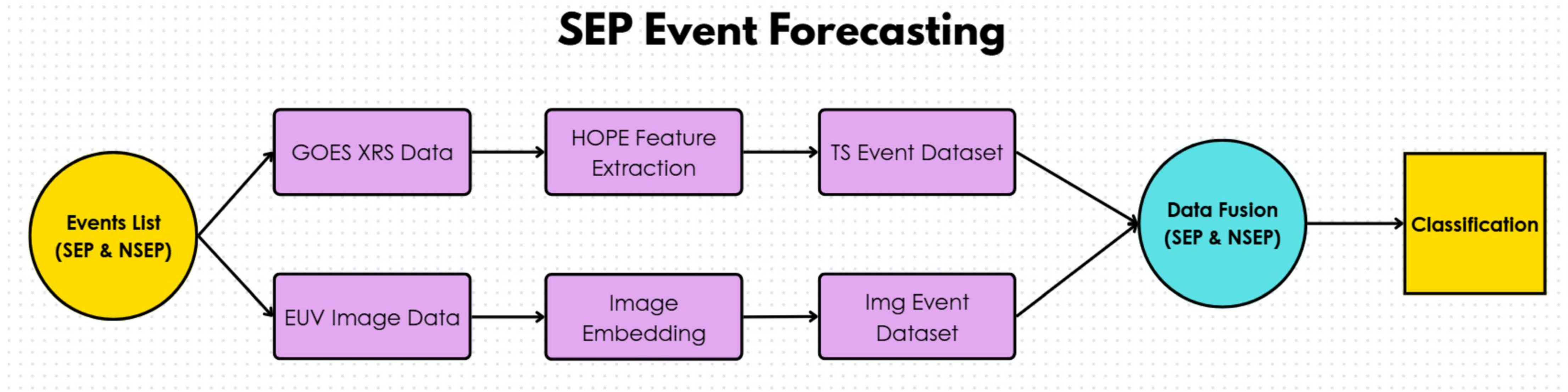
- **HOPE-derived time-series features** (in-situ measurements over time)
- **EUV solar images** (visual signatures of solar activity)

This project uses machine learning to combine this information. The HOPE features provide time-based data while the EUV images provide image-based data. They are able to allow the model to learn the changes over time. A multimodal model (one that combines multiple types of data) will be used with a BiLSTM and a CNN.

Research question: Can combining HOPE time-series signals with EUV image information improve SEP vs. non-SEP classification compared to a single-source model?

Hypothesis: Solar events that produce SEP events will show early warning patterns with **HOPE features** and **EUV images**. Therefore, combining GOES X-ray features with EUV image models will improve the model's ability to forecast these events.

Methods



Data + Labels

Build labeled time windows for SEP and NSEP periods
Match all inputs to consistent timestamps

HOPE Time-Series Model

Preprocessing: cleaning, timestamp alignment, missing-value handling, scaling/normalization
Windowing: fixed-length sequences (e.g., past [X hours] to predict [Y-hour] label)
Model: [LSTM/GRU/TCN/Transformer] trained for SEP classification

EUV Image Model

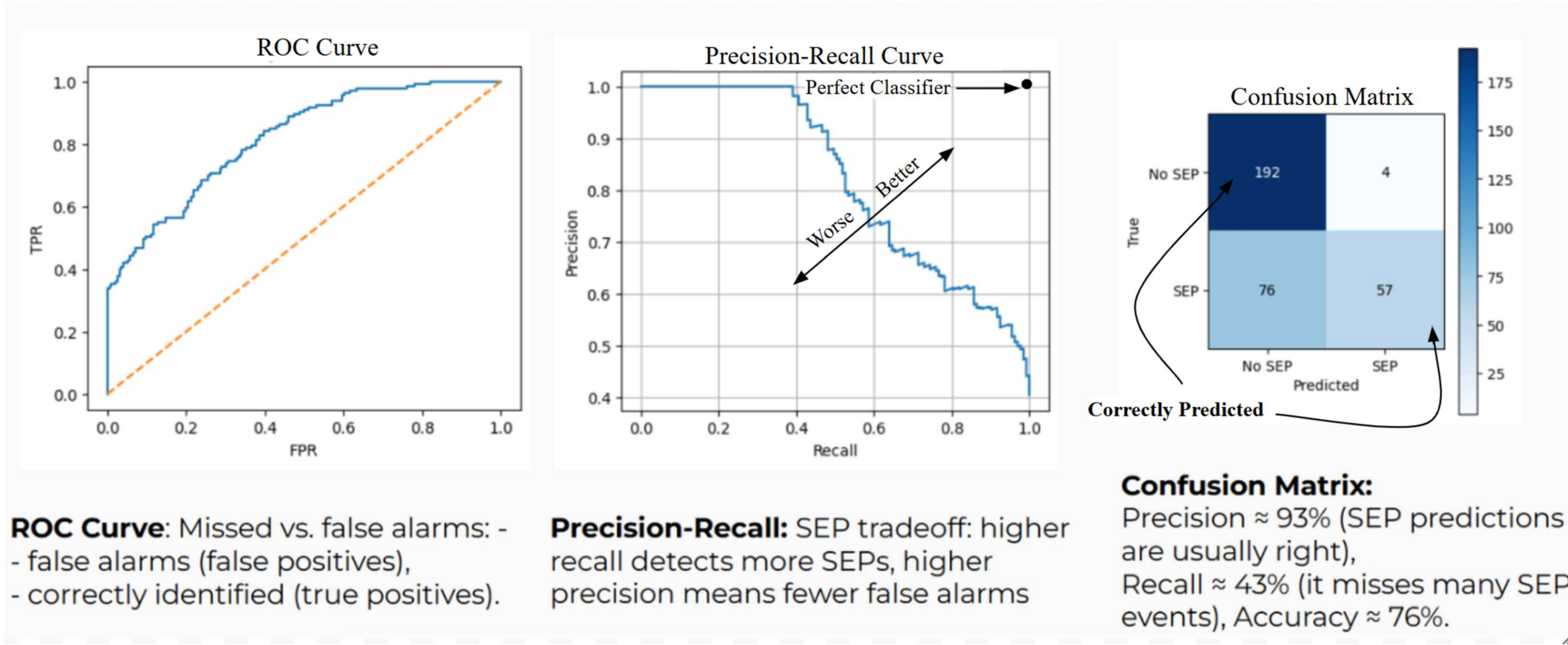
Preprocessing: timestamp matching, resizing/cropping, normalization, quality
Model: CNN backbone [e.g., ResNet-like] trained on EUV images aligned to the same label windows

Fusion (Combined Model)

Combined HOPE + EUV embeddings (concatenation or attention fusion)
Final classification head outputs SEP probability
Trained with imbalance handling: [class weights / balanced sampling]
Evaluation: time-aware split (train earlier, test later) to reduce leakage

Findings

The model is very cautious: it produces few false alarms, but it misses many SEP events (low recall). This shows the importance of improving SEP detection and handling class imbalance.



ROC Curve: Missed vs. false alarms: - false alarms (false positives), - correctly identified (true positives).

Precision-Recall: SEP tradeoff: higher recall detects more SEPs, higher precision means fewer false alarms

Confusion Matrix:
Precision $\approx 93\%$ (SEP predictions are usually right),
Recall $\approx 43\%$ (it misses many SEP events), Accuracy $\approx 76\%$.

Discussion

The model can learn meaningful patterns, but the results suggest the model may be biased toward predicting NSEP due to class imbalance. In SEP forecasting, missing true SEP events can be costly, so improving recall (catching more SEPs) is a priority.

Limitations:

- Class imbalance (SEP events are less common, which can cause the model to under-predict SEP with many false negatives)
- Threshold choice: (Using a default cutoff (0.5) may not match the real-world goal of catching more SEPs)
- Data complexity: (EUV images require much preprocessing and careful timestamp pairing to labels)
- Generalization: (Performance may be different across solar cycles or different event definitions, more testing is needed)

Future Work:

- Add more SEP events from multiple years of GOES data
- Include non-SEP flare events to help the model learn the difference between SEP and non-SEP indicators
- Use multiple EUV image wavelengths such as 171 Å, 193 Å, and 304 Å to capture different layers of the solar atmosphere
- Add magnetic-field features from active regions such as sunspot area and magnetic complexity.

References

Abad-Alcaraz, V., et al. (2025). Multimodal deep learning for solar radiation forecasting. *Solar Energy*, 283, 124–137. <https://doi.org/10.1016/j.apenergy.2025.126061>

Hosseinzadeh, P., Filali Boubrahimi, S., & Hamdi, S. M. (2025). An end-to-end ensemble machine learning approach for predicting high-impact solar energetic particle events using multimodal data. *The Astrophysical Journal Supplement Series*, 277(2), 34. <https://doi.org/10.3847/1538-4365/adb1c4>

Hosseinzadeh, P., et al. (2024). Toward enhanced prediction of high-impact solar energetic particle events using multimodal time-series data fusion models. *Space Weather*, 22(11). <https://doi.org/10.1029/2024SW003982>

Massachusetts Institute of Technology. (2006). Introduction to Plasma Physics I (Course 22.611J). MIT OpenCourseWare. Retrieved from <https://ocw.mit.edu/courses/22-611j-introduction-to-plasma-physics-i-fall-2006/>

Palmerio, E., et al. (2024). Improved modelling of SEP event onset within the WSA–Enlil+Cone framework. *Journal of Space Weather and Space Climate*, 14, 1. <https://doi.org/10.1051/swsc/2024035>

Telikicherla, A., Woods, T. N., & Schwab, B. D. (2025). Improving solar flare nowcasting with the Hot Onset Precursor Event (HOPE) technique. *Astronomy & Astrophysics* (in press). Preprint at arXiv:2509.05234. <https://arxiv.org/abs/2509.05234>

